

Hridoy Sankar Dutta

hridoysankardutta@gmail.com, +447469674172

UK Permanent Resident (No Visa Sponsorship Required)

PROFESSIONAL HIGHLIGHTS

- PhD researcher with 5+ years of experience working at the intersection of cybersecurity, data science, and large-scale online systems, focusing on measurement and analysis of complex digital ecosystems.
- Extensive experience designing and managing large-scale datasets and automated data pipelines for scientific research, including the development of widely used open-source datasets such as [YT-30M](#) and [ABOME](#). The datasets have received 1500+ downloads and have been adopted by industry (e.g., ActiveFence) and academia for cybercrime analysis, social media intelligence and large-scale online safety research.
- Proven ability to conduct interdisciplinary research and collaborate across domains, with academic experience at the University of Cambridge and Deakin University, translating complex technical insights to diverse research and engineering teams.
- Published impactful research in leading peer-reviewed conferences and journals including TIFS, ASIACCS, WiSec, eCrime, TIST, ICWSM, CIKM, and WSDM, demonstrating strong experience in rigorous data-driven scientific research.

EXPERIENCE & EDUCATION

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| Lecturer in Cybersecurity, Deakin University | July 2024 - Present |
| Research Associate, University of Cambridge
<i>Cambridge Cybercrime Centre</i>
<i>University of Cambridge</i> | June 2021 - June 2024 |
| Research Assistant, University of Cambridge
<i>Cambridge Cybercrime Centre</i>
<i>University of Cambridge</i> | February 2021 - June 2021 |
| PhD in Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIIT-D) | August 2017 - September 2021 |
- Thesis Title: *Detecting and reasoning collusive activities in online media platforms.*

SELECTED PUBLICATIONS

- Google Scholar: <https://scholar.google.co.in/citations?user=4UrQjwUAAAAJ>.
- **[InterSpeech-26]** Kamel Kamel, **Hridoy Sankar Dutta**, Keshav Sood and Sunil Aryal. Spectral Masking and Interpolation Attack (SMIA): A Black-box Adversarial Attack against Voice Authentication and Anti-Spoofing Systems. *27th Conference of the International Speech Communication Association (InterSpeech 2026)*, September 27 – October 1, 2026.
 - **[SIGIR-26]** **Hridoy Sankar Dutta**, Biswadeep Khan and Amit Nanavati. GONets: A First-Look Into GitHub Organisation Networks. *49th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2026)*, July 20 – 24, 2026.
 - **[WiSec-26]** Luis Adán Saavedra del Toro, **Hridoy Sankar Dutta**, Alastair Beresford and Alice Hutchings. iOSZoo: A Large-Scale Study of Third-Party iOS App Markets. *ACM Conference on Security and Privacy in Wireless and Mobile Networks (ACM WiSec 2026)*, June 30 – July 3, 2026, Saarbrücken, Germany.
 - **[CIKM-25]** **Hridoy Sankar Dutta** and Biswadeep Khan. YTCommentVerse: A Multi-Category Multi-Lingual YouTube Comment Corpus. *34th ACM Conference on Information and Knowledge Management (CIKM 2025)*, Nov 10 – 14, 2025.
 - **[ASIACCS-25]** Luis Adán Saavedra del Toro, **Hridoy Sankar Dutta**, Alastair Beresford and Alice Hutchings. App-solutely Modded: The rift between modded app market operators and original developers. *ACM ASIA Conference on Computer and Communications Security (ACM ASIACCS 2025)*, August 25 – 29, 2025, Ha Noi, Vietnam.
 - **[eCrime-24]** Luis Adán Saavedra del Toro, **Hridoy Sankar Dutta**, Alastair Beresford and Alice Hutchings. ModZoo: A Large-Scale Study of Modded Android Apps and their Markets, *APWG Symposium on Electronic Crime Research (eCrime 2024)*, September 24 – 26, 2024, Boston, Massachusetts, USA.
 - **[IEEE TIFS]** **Hridoy Sankar Dutta**, Vishal Raj Dutta, Aditya Adhikary and Tanmoy Chakraborty. HawkesEye: Detecting fake retweeters using Hawkes process and topic modeling. *IEEE Transac-*

	<i>tions on Information Forensics and Security (TIFS)</i> 15 (2020): 2667-2678.															
	• [ICWSM-22] Hridoy Sankar Dutta, Nirav Diwan, Tanmoy Chakraborty. Weakening the Inner Strength: Spotting Core Collusive Users in YouTube Blackmarket Network, <i>16th International AAAI Conference on Web and Social Media (ICWSM)</i>. vol. 16, pp. 147-158. 2022. • [IEEE TIFS] Hridoy Sankar Dutta and Tanmoy Chakraborty. Blackmarket-driven Collusion among Retweeters – Analysis, Detection and Characterization. <i>IEEE Transactions on Information Forensics and Security (TIFS)</i> 15 (2019): 1935-1944.															
PATENT	• Sujoy Saha, Prithviraj Pramanik, Partha Sarathi Paul, Subrata Nandi, Kingshuk De, Bishak Chandra Ghosh, Hridoy Sankar Dutta, Tamal Mondal, Krishnendu Hazra, Indrajit Bhattacharya, Sandip Chakraborty, An end-to-end system for offline localized crisis mapping and method of operation thereof, Indian Patent, Application No: 201931050685.															
DATASETS	• YT-30M: A multi-lingual multi-category dataset of YouTube comments [CIKM'25]. <i>HuggingFace downloads: 691, Industry adoption: ActiveFence (Alice)</i> • ABOME: A Multi-platform Data Repository of Artificially Boosted Online Media Entities. [ICWSM'21] <i>Zenodo downloads: 711</i> • ModZoo: A large-scale dataset of 159K modded Android apps through longitudinal measurement of 13 major modded app marketplaces. [eCrime'24] • iOSModZoo: Built large-scale datasets of 40K modded iOS apps and 77K App Store counterparts through 11-month monitoring of 9 modded markets, enriched with metadata from 1.02M official iOS apps. [WiSec'26] • GONets: A first-look into GitHub Organisation Networks [Under submission at SIGIR'26]															
OPEN-SOURCE CONTRIBUTIONS	See GitHub profile at https://tinyurl.com/hsd-github. • The YouTube Scraper - Tool to download YouTube video descriptions and comments without using the YouTube API. (<i>160+ stars, 25+ forks</i>) • Awesome Twitter Tools - A curated list of Twitter tools and resources. (<i>200+ stars, 25+ forks</i>)															
INTERESTS AND SKILLS	Interests: Data-driven cybersecurity, Cybercrime, Social network safety, LLM Safety. Operating systems: GNU/Linux; Tools: Python, SQL, Deep Learning (Graphs), GitHub.															
GRANTS	<table><tr><td>Deakin University Peer-review ECR Support Scheme</td><td>4250 AUD</td></tr><tr><td>Deakin University SIT Minor Equipment & Expenses Scheme 2025</td><td>3500 AUD</td></tr><tr><td>Deakin University Startup Grant (PI)</td><td>INR 10000 AUD</td></tr><tr><td>Deakin Cyber Collaboration Grants Scheme 2025/2026 (Co-PI)</td><td>10000 AUD</td></tr><tr><td colspan="2"><i>Deakin University</i></td></tr><tr><td>OpenAI Researcher Access Program (PI)</td><td>1000 USD</td></tr><tr><td>Cohere Labs Catalyst Grant Program</td><td>500 USD</td></tr></table>		Deakin University Peer-review ECR Support Scheme	4250 AUD	Deakin University SIT Minor Equipment & Expenses Scheme 2025	3500 AUD	Deakin University Startup Grant (PI)	INR 10000 AUD	Deakin Cyber Collaboration Grants Scheme 2025/2026 (Co-PI)	10000 AUD	<i>Deakin University</i>		OpenAI Researcher Access Program (PI)	1000 USD	Cohere Labs Catalyst Grant Program	500 USD
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OTHER EXPERIENCES	Co-founder, Assam AI Initiative (AII) <table><tr><td><i>A non-profit initiative to promote various scopes of AI</i></td><td>Feb 2021 - Present</td></tr><tr><td>Visiting Researcher, TU Munich, Germany</td><td>July 2023</td></tr><tr><td>Visiting Researcher, Cardiff University, UK</td><td>March 2019, March 2020</td></tr><tr><td>Short-Term Scholar, George Mason University, USA</td><td>June 2019 - August 2019</td></tr></table>		<i>A non-profit initiative to promote various scopes of AI</i>	Feb 2021 - Present	Visiting Researcher, TU Munich, Germany	July 2023	Visiting Researcher, Cardiff University, UK	March 2019, March 2020	Short-Term Scholar, George Mason University, USA	June 2019 - August 2019						
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ACHIEVEMENTS AND AWARDS	• Rising Star in Data Science by the Center for Data and Computing (CDAC), University of Chicago. • Wiseman Prize 2023, Department of Computer Science and Technology, University of Cambridge. • DAAD AInet fellowship 2023. • SEBE Teaching Award, Deakin University. • ICWSM 2021 scholarship for Underrepresented Groups Program. • 3rd prize at Ninth IDRBT Doctoral Colloquim IDC 2019.															